

Replicating Summary Employment Statistics from Jardim et al. (2022)

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1 The Original Paper

Jardim, Ekaterina, Mark C. Long, Robert Plotnick, Emma van Inwegen, Jacob Vigdor, and Hilary Wething. 2022. “Minimum Wage Increases and Low-Wage Employment: Evidence from Seattle.” *American Economic Journal: Economic Policy* 14(2): 263–314.

The paper evaluates the employment and wage effects of Seattle’s Minimum Wage Ordinance, which raised the minimum wage from \$9.47 to \$11 per hour in April 2015 and to \$13 per hour in January 2016. The authors use administrative unemployment insurance records from Washington State to directly observe hourly wages for the entire workforce, allowing them to focus precisely on low-wage jobs across all industries. Their empirical strategy combines simple first-difference analysis with synthetic control and interactive fixed effects estimators to isolate the causal effect of the ordinance from Seattle’s broader economic boom.

2 The Result of Interest

I replicate the summary employment statistics reported in Tables 5a, 5b, and 5c of Jardim et al. (2022), which document changes in the number of jobs, total hours worked, and average wages for Seattle’s locatable establishments between 2014 Q2 and 2016 Q2. These three tables together show that low-wage employment contracted in Seattle even as overall employment expanded dramatically during the same period, providing the descriptive foundation for the paper’s causal synthetic control analysis that follows.

3 Data and Methods

The data come from the publicly available replication package deposited by Jardim et al. (2022) on the ICPSR repository (project 133921). I use the file `region_level_cumulative.dta`, a PUMA-by-quarter aggregated dataset constructed from confidential unemployment insurance records. Each row represents one Public Use Microdata Area in Washington State for one quarter, covering 2014 Q2 through 2016 Q3.

The preprocessing script reads the raw data file, filters to the relevant quarter range, flags Seattle PUMAs using the region indicator variable, and aggregates across PUMAs by

summing within each Seattle-by-quarter cell. Average hourly wage rates are derived by dividing total quarterly payroll by total quarterly hours. The cleaned dataset is saved to `temp/clean_data.csv`.

The analysis script reads the cleaned data, extracts the baseline (2014 Q2) and endpoint (2016 Q2) observations for Seattle, and computes the absolute and percentage changes in jobs, hours, and wages for jobs paying under \$19 per hour. These values are written directly to a LaTeX table fragment in `output/tables/main_result.tex`, which is included below.

4 Replication Results

Table 1: Replication of Summary Employment Statistics, from Jardim et al. (2022)

	2014 Q2	2016 Q2	Change
Jobs paying < \$19/hr	90757	89188	-1.7%
Hours worked < \$19/hr	36450720	35466512	-2.7%
Avg. wage < \$19/hr	\$14.19	\$15.00	5.7%

Our replication results are shown in the table above. All three measures; number of jobs, hours worked, and average wage; match the published paper exactly. Jobs paying under \$19 per hour declined between 2014 Q2 and 2016 Q2, total hours worked in such jobs fell over the same period, and average wages rose. The perfect match between our results and the published paper confirms that the replication package provided by the authors on ICPSR faithfully reproduces the summary statistics reported in Tables 5a, 5b, and 5c of the published version.

5 What I Learned

The most straightforward part of this replication was the analysis itself. Once the data were properly loaded and aggregated, the summary statistics matched the published paper exactly across all three outcomes — jobs, hours, and wages. This confirmed that the replication package provided by the authors faithfully reproduces the published results.

The harder lesson concerns the limits of what can be replicated publicly. The most important results in the paper, the synthetic control estimates of the causal effect of the minimum wage, require confidential unemployment insurance microdata that cannot be shared publicly. As an outside replicator, I could only verify the descriptive statistics, not the core causal findings. This taught me that even a high quality replication package has boundaries, and that data access is often the binding constraint in replication exercises.

References

- [1] Jardim, Ekaterina, Mark C. Long, Robert Plotnick, Emma van Inwegen, Jacob Vigdor, and Hilary Wething. 2022. “Minimum Wage Increases and Low-Wage Employment: Evidence from Seattle.” *American Economic Journal: Economic Policy* 14(2): 263–314.